वित्तीय सेवाएं विभाग
DEPARTMENT OF
FINANCIAL SERVICES

PSBs Hackathon Series 2026

(An Initiative of Government of India, Ministry of Finance , Department of Financial Services)

यूनियन बैंक ऑफ इंडिया Union Bank of India

PRESENTS

IDEA 2.0

INNOVATION | DEVELOPMENT | EXECUTION | ADVANCEMENT

A i - C S P A R C

AI Powered Innovation for Customer: Security, Privacy, Assurance, Reliance, Convenience

Digital Trust | Ease of doing Business

Promoting Innovation and Fostering Collaboration

SOMAIYA VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering

(formerly K J Somaiya College of Engineering)

PROBLEM STATEMENT & TEAM DETAILS

TEAM NAME (AS SUBMITTED ON PORTAL)		ByteBrains	
PROBLEM STATEMENT TITLE		Predictive Customer Outreach for Connection and Retention	
	NAME	**GENDER**	**AREA OF EXPERTISE**
Member 1	Vrushika K Panchal	Female	Python, Machine Learning, Data Science, Computer Vision, Data Analysis, Tableau, OpenCV, Deep Learning
Member 2	Movin D'Souza	Male	Python, Next.js, FastAPI, Backend Development, Frontend Development, LangChain, LangGraph, Agentic AI Systems
Member 3	Yash Dhasal	Male	Machine Learning, Deep Learning, Generative AI, Python, React, JavaScript, Applied AI Systems Development
Member 4	Nabeel Ahmad	Male	Operations Management, Project Coordination, Research, Blockchain Technology

IDEA TITLE

IDEA / SOLUTION

RetentionOS is an **AI-driven predictive customer outreach platform** that uses **churn prediction, causal uplift modeling,** and **profit-aware decision intelligence** to automatically **identify persuadable customers** and **execute personalized retention strategies.**

Main Points of Interest -

- **Built for Customer-Driven Industries:** Banks, telecom, subscription platforms, and digital services to detect churn early and protect customer lifetime value.
- **Automated Intelligent Retention:** Identifies high-value at-risk customers, selects the best intervention strategy, and executes outreach across multiple channels.

PROBLEM RESOLUTION

Early Risk Detection & Customer Prioritization: Identify high-value customers showing early disengagement and focus on those most likely to be retained.

Causal Uplift Segmentation & AI-Guided Actions: Segment customers into stay, leave, and persuadable groups, then select the most effective and resource-efficient intervention for each customer.

UNIQUE VALUE PROPOSITION (UVP)

RetentionOS moves **beyond traditional churn prediction** by **identifying persuadable customers** and **executing profit-optimized retention actions** using causal **uplift intelligence** and **AI-driven strategy decisions.**

OUTLINE OF UNIQUE & INNOVATIVE SOLUTION

Key Features of RetentionOS

An intelligent retention engine that transforms churn prediction into profit-driven customer persuasion.

Financial & Risk Intelligence

LTV-based customer eligibility filtering with ML-driven churn risk detection.

Causal Uplift Intelligence

Identifies customers who can actually be influenced by retention interventions.

AI Strategy Decision Engine

Agent-based reasoning selects the best retention strategy using churn risk, uplift score, and customer value.

Personalized Multi-Channel Outreach

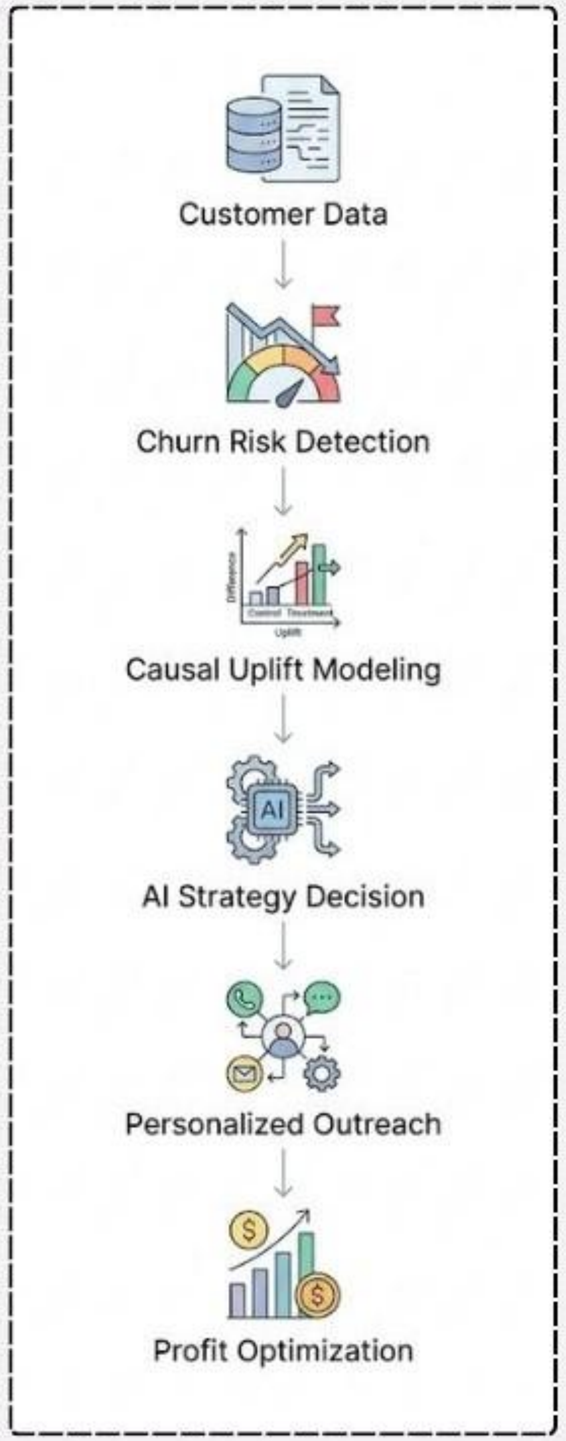
Automates targeted engagement via email, SMS, app notifications, or calls.

Context & Network Intelligence

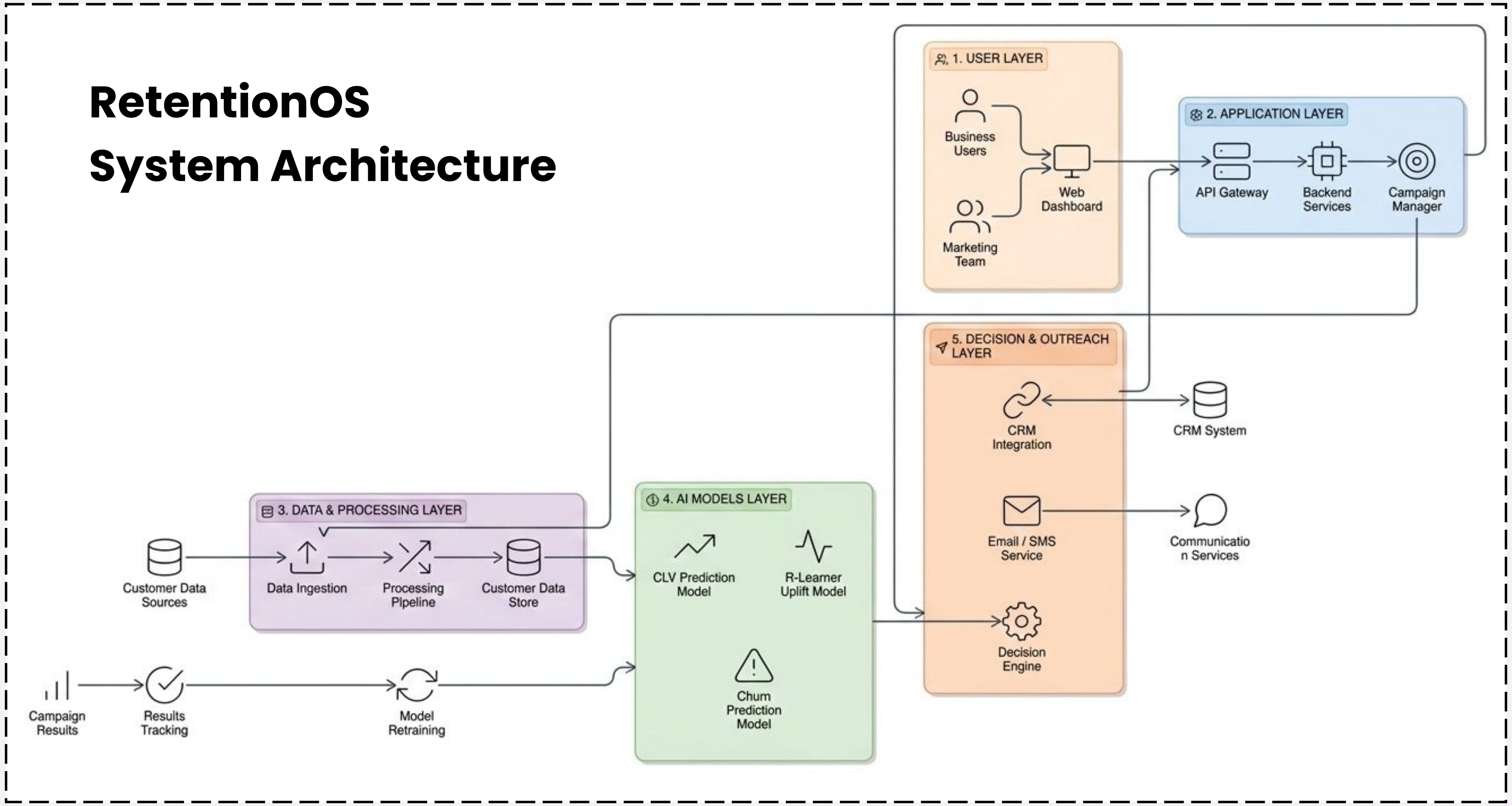
Uses knowledge graphs and influence modeling to detect hidden churn risks.

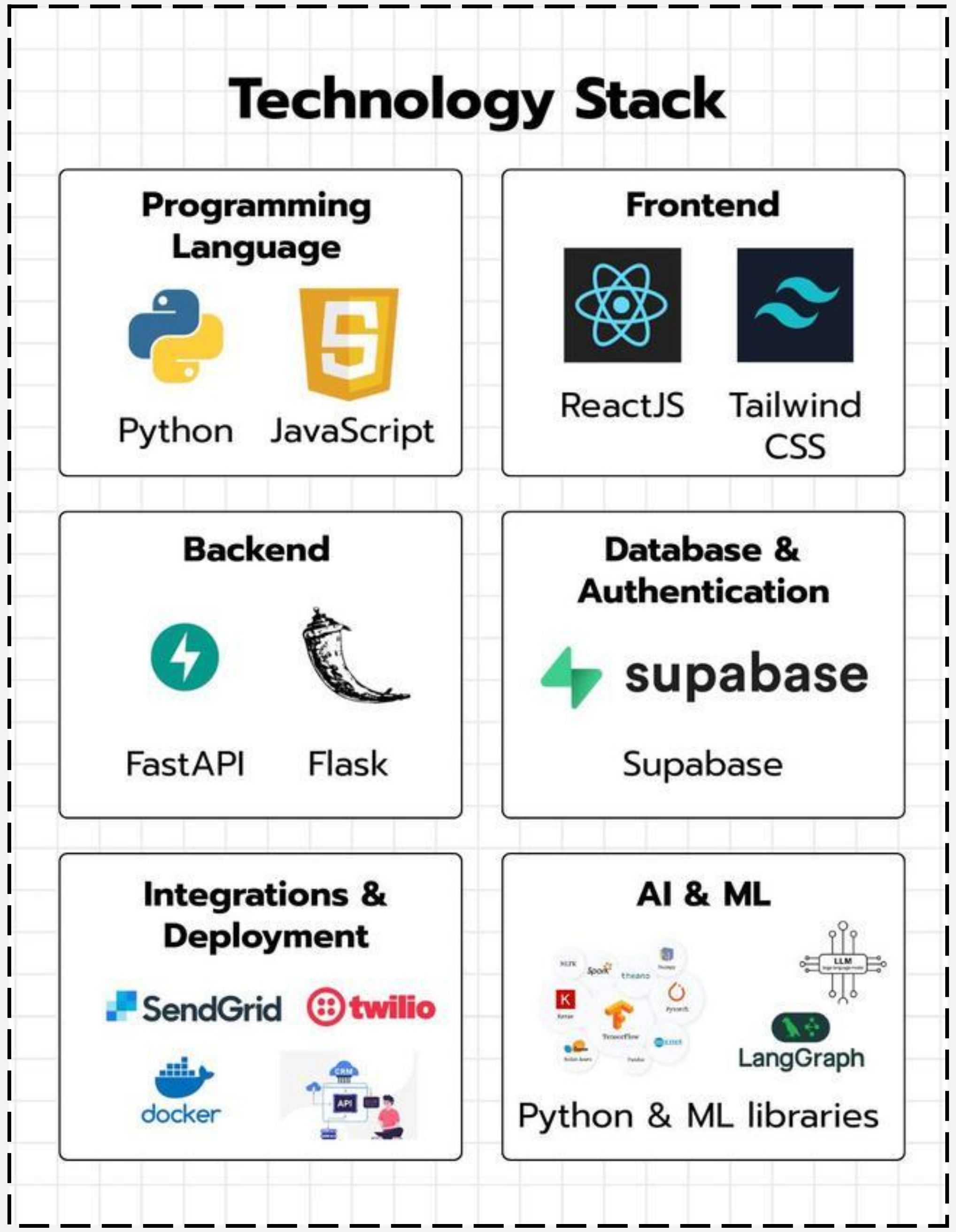
Profit Guardrails & Market Awareness

Ensures ROI-positive actions using policy controls, competitor monitoring, and strategy simulation.



TECHNICAL APPROACH

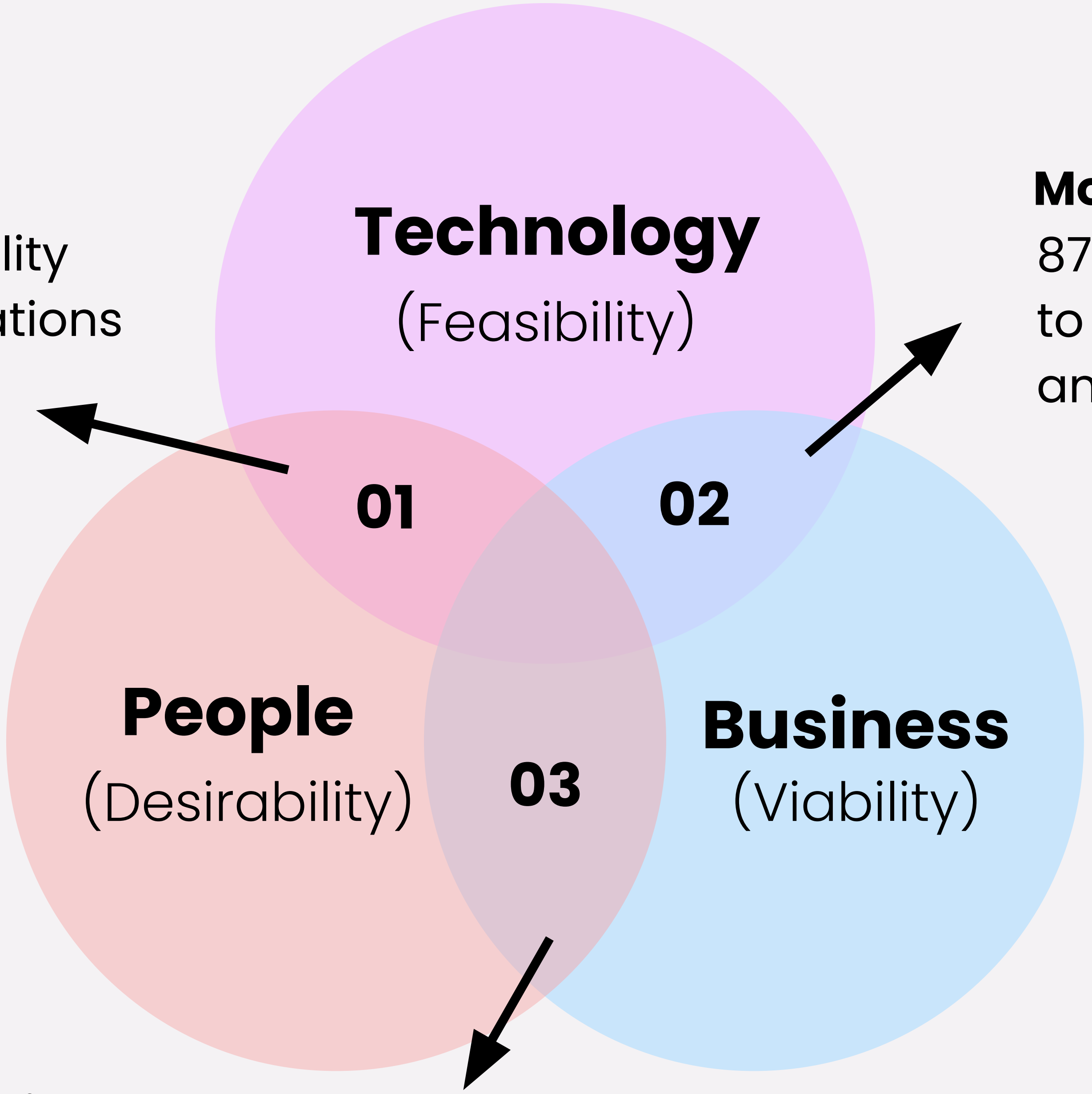




A modular retention intelligence architecture supported by a modern tech stack for data processing, causal ML modeling, AI decision workflows, and automated outreach.

FEASIBILITY & VIABILITY

Data Risk
Poor data quality costs organizations up to $12.9M annually.



Model Risk
87% of AI projects fail due to poor model deployment and accuracy issues.

Trust Risk
63% of customers are concerned about how companies use their data.

01 — **71% of customers expect personalized interactions, and better experience directly improves loyalty.**

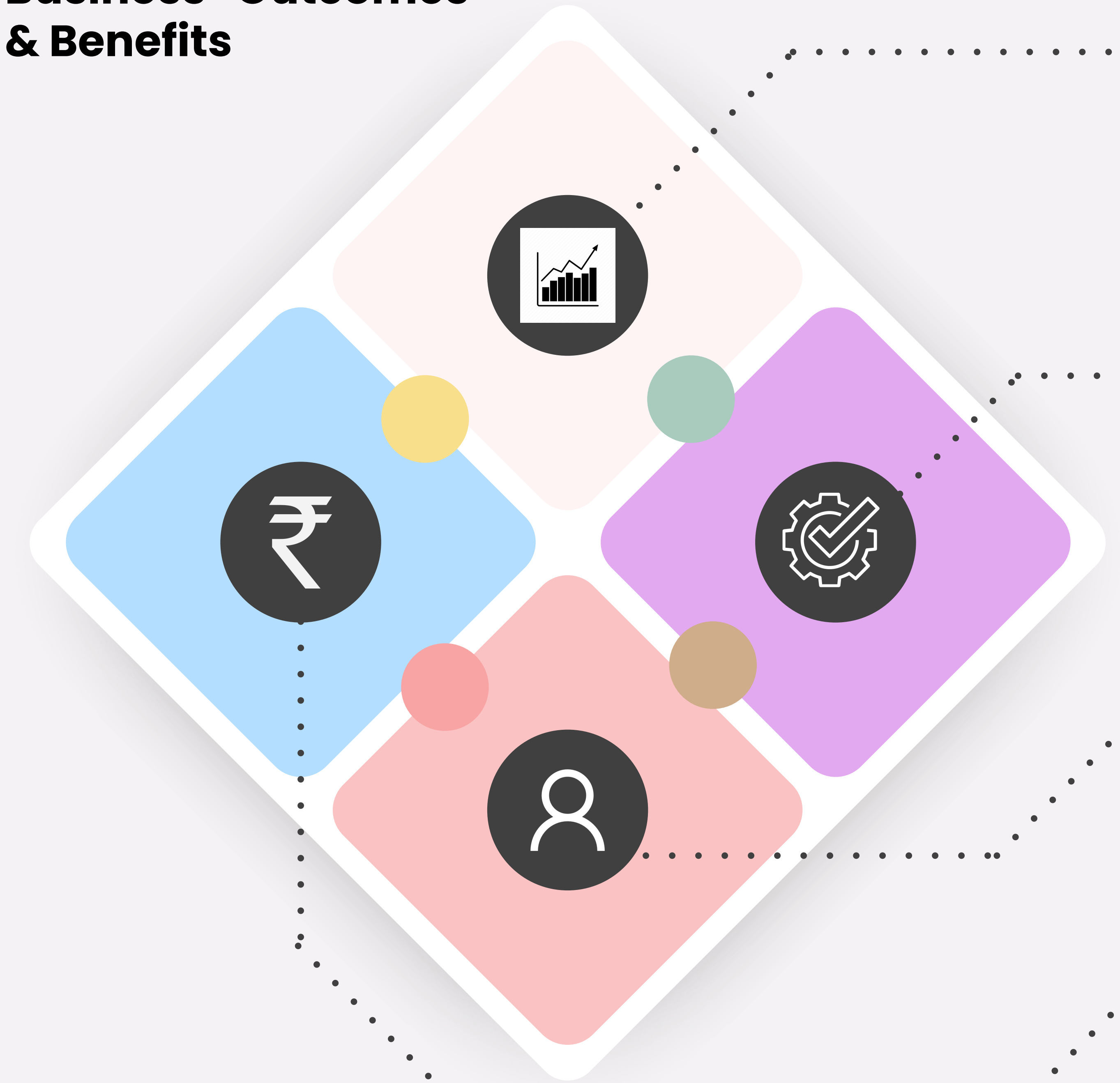
02 — **AI-driven retention systems can reduce churn by up to 15–20% and improve targeting accuracy significantly.**

03 — **Reducing churn by just 5% can increase profits by 25% to 95%.**

Mitigates risks through data validation, continuous model learning, secure systems, and human-in-the-loop decision support.

IMPACT & BENEFITS

Business Outcomes & Benefits



01 **Business Growth & Profitability**

Increases customer lifetime value by targeting high-value, persuadable customers with profit-optimized retention strategies.

02 **Intelligent Decision Automation**

Enables AI-driven decisions using churn prediction, causal uplift modeling, and automated strategy selection.

03 **Enhanced Customer Experience**

Delivers personalized and timely interventions, improving engagement, satisfaction, and long-term loyalty.

04 **Cost & Resource Optimization**

Minimizes wasted incentives by focusing only on customers with high retention impact and ROI potential.

Key Impact Metrics

15–20%
Retention Improvement

25–95%
Profit Increase

Reduced
Marketing Waste

Higher
Customer Lifetime Value

Driving measurable retention, profit, and efficiency gains.

BUSINESS MODEL

Key Partners
- CRM & marketing automation platforms
- Cloud infrastructure providers
- Data analytics vendors
- Enterprise system integrators

Key Activities
- AI/ML model development
- Data integration & processing
- Automated retention campaigns

Key Resources
- AI / ML models & data pipelines
- Customer behavioral datasets
- Cloud infrastructure

Value Proportions
- AI-driven early churn detection
- Identify persuadable customers using causal ML
- Profit-optimized retention strategies
- Automated personalized outreach

Customer Relationships
- Enterprise onboarding & support
- Dedicated success managers
- AI model improvements

Channels
- Technology partnerships / CRM integrations
- Direct enterprise sales & Cloud SaaS platform
- Industry conferences & fintech events

Customer Segments
- Banks & Financial Institutions
- Telecom Providers
- Subscription & SaaS Platforms
- E-commerce & Digital Services

Cost Structure
- Cloud computing & infrastructure
- AI/ML development & maintenance
- Data acquisition & processing
- Sales, marketing & enterprise support

Revenue Streams
- SaaS subscription licensing
- Enterprise implementation fees
- Usage-based pricing (campaign volume)
- Premium analytics & insights modules

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RESEARCH & REFERENCES

1. Causal Intelligence & R-Learner (Section 8.4):
Enables accurate estimation of treatment effects to identify customers who are truly persuadable through targeted interventions. Link: https://arxiv.org/abs/1712.04912

2. Uplift Modeling & Rubin Causal Model (Section 6):
Differentiates customers based on their likelihood to respond, forming the foundation for precise and targeted retention strategies. Link: https://proceedings.mlr.press/v67/gutierrez17a/gutierrez17a.pdf

3. Agentic Multi-Agent Decision Systems (Section 9):
Enables intelligent, multi-step decision workflows for automated and adaptive retention strategy execution.
Link: https://arxiv.org/abs/2502.10978

4. Graph-Based Customer Influence & Churn (Section 8.8):
Identifies influential customers and network effects, helping prevent cascading churn through early intervention.
Link:https://www.researchgate.net/publication/286426990_A_Graph-Based_Churn_Prediction_Model_for_Mobile_Telecom_Networks

Idea Summary:
https://drive.google.com/file/d/1K3D0PNvGSKj8dESTgathXDEM9V5eJuJW/view?usp=sharing